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(22SCSE1012097) (13-Mar-2026)

Practical Test: Prompt Engineering (Beginner Level)

Duration: 45–60 minutes

Tool Allowed: Any LLM (ChatGPT, Gemini, Claude, etc.)

Section A: Prompt Writing (10 Marks)

Question 1

Write a **clear and structured prompt** that asks an AI model to:

Explain **Machine Learning** to a **class 10 student** using **simple language and real-life examples**.

Expected Skills Tested

- Clear instruction
- Audience specification
- Output clarity

Answer 1:-

Act as a knowledgeable teacher.

Explain the concept of Machine Learning to a Class 10 student.

Use simple and easy-to-understand language and avoid complex technical terms.

Include 2–3 real-life examples (such as YouTube recommendations, voice assistants, or spam email filters) to make the concept clearer.

Structure the explanation in the following way:

1. Definition of Machine Learning
2. How it works in simple terms
3. Real-life examples

Keep the explanation clear, short, and student-friendly.

Question 2

Write a prompt that asks an AI model to generate:

- **5 quiz questions**
- Topic: **Artificial Intelligence**
- Difficulty: **Easy**
- Include **answers**

Answer 2:-

Act as an experienced quiz creator and AI educator.

Create a beginner-level quiz on the topic Artificial Intelligence.

Requirements:

- Generate exactly 5 quiz questions.
- Difficulty level: Easy
- The questions should test basic understanding of Artificial Intelligence concepts.
- Use simple and clear language suitable for beginners.
- Prefer multiple-choice questions with four options (A, B, C, D).

Output Format:

For each question include:

1. Question
2. Four options (A, B, C, D)
3. Correct answer clearly indicated

Ensure the quiz is short, clear, educational, and easy to understand.

Section B: Prompt Improvement (10 Marks)

Below is a **poor prompt**:

"Tell me about AI."

Task:

Rewrite this prompt to make it **more effective and specific**.

Your improved prompt must include:

- Target audience
- Output format
- Examples or explanation

Answer:-

Act as an experienced Artificial Intelligence educator.

Explain the concept of Artificial Intelligence (AI) to Class 10 students who are new to the topic.

Requirements:

- Use simple, clear, and beginner-friendly language.
- Avoid complex technical terms.
- Explain the concept in a way that students can easily understand.

Include the following in your explanation:

1. A clear definition of Artificial Intelligence
2. A simple explanation of how AI works
3. 2–3 real-life examples such as voice assistants, recommendation systems, or self-driving cars

Output Format:

- Definition
 - Explanation
 - Real-life Examples
 - Short Conclusion summarizing the concept
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Section C: Role-based Prompting (10 Marks)

Write a prompt that asks the AI to:

- Act as a **Data Science teacher**
- Explain **Cosine Similarity**
- Include:
 - Formula
 - Simple example
 - Python code snippet

Answer:-

You are an experienced Data Science teacher explaining concepts to undergraduate students.

Explain the concept of Cosine Similarity in a clear and structured way suitable for beginners in data science.

Requirements:

- Begin with a brief and intuitive definition of Cosine Similarity.
- Explain why Cosine Similarity is used in data science, especially in areas like text similarity, recommendation systems, or document comparison.
- Include the mathematical formula and briefly describe what each variable represents.
- Provide a simple numerical example using two small vectors and explain the calculation step-by-step.
- Include a Python code snippet that calculates Cosine Similarity between two vectors using common libraries such as NumPy or scikit-learn.

Output Format:

1. Concept Overview
 2. Mathematical Formula
 3. Step-by-step Numerical Example
 4. Python Code Snippet
 5. Short Practical Use Case
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Section D: Structured Output Prompt (10 Marks)

Write a prompt that asks the AI to generate a **table** containing:

Algorithm	Use Case	Example
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for **5 Machine Learning algorithms**.

Answer:-

Act as an experienced Machine Learning instructor and AI researcher.

Generate information about five widely used Machine Learning algorithms and present the information in a well-structured table suitable for students learning introductory Machine Learning.

Requirements:

- Select five commonly used Machine Learning algorithms (e.g., Decision Tree, Linear Regression, K-Nearest Neighbors, Support Vector Machine, Random Forest).
- For each algorithm, provide:
 - The name of the algorithm
 - Its primary use case

- A real-world example where the algorithm can be applied

Output Format:

Create a structured table with the following columns:

| Algorithm | Use Case | Example |

Guidelines:

- Keep explanations clear, concise, and beginner-friendly.
- Ensure the examples represent practical real-world applications of each algorithm.

Section E: Prompt Debugging (10 Marks)

An AI produced a **very long and complex answer** when asked a question.

Write a prompt that instructs the AI to:

- Give the answer in **5 bullet points**
- Use **simple language**
- Limit response to **100 words**

Answer:-

You are an expert communication coach who specializes in simplifying complex information.

Your task is to answer the given question in a **clear, concise, and beginner-friendly manner**.

Instructions:

- Use **simple and easy-to-understand language** suitable for a general audience.
- Avoid technical jargon and overly complex explanations.
- Present the response in **exactly 5 bullet points**.
- Each bullet point should contain **one key idea only**.
- Keep each bullet point **short and informative**.

Constraints:

- The **total response must not exceed 100 words**.

- Do **not** include paragraphs or extra explanations outside the bullet points.
- Focus only on the **most important information**.

Output Format:

- Bullet Point 1
- Bullet Point 2
- Bullet Point 3
- Bullet Point 4
- Bullet Point 5

Bonus Question (Optional – 5 Marks)

Write a prompt that asks an AI to generate a **Python program that performs sentiment analysis on text**.

Answer:-

Act as an experienced Python developer and Data Science instructor.

Generate a Python program that performs sentiment analysis on text.

Requirements:

- Use a common Python library such as TextBlob, NLTK, or scikit-learn.
- The program should accept a text input from the user.
- Analyze the sentiment of the text and classify it as Positive, Negative, or Neutral.
- Include clear comments in the code explaining each step.

Output Format:

1. Brief explanation of the approach used for sentiment analysis
 2. Complete Python code snippet
 3. Example input and output demonstrating how the program works
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Evaluation Rubric

Criteria	Marks
Clarity of prompt	10
Use of constraints	10
Structured prompting	10
Role-based prompting	10
Output formatting	10